

CONSTRUCTPRO ERP

ABOUT THE SYSTEM

Construction Project Management & Enterprise Resource Planning System

Version 1.0

System Overview and Functional Profile

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1. SYSTEM OVERVIEW

ConstructPro ERP is a construction-focused Enterprise Resource Planning and Project Management system designed to bring the operational activities of a construction organization into one integrated platform.

The system is designed around projects, procurement, inventory, warehouses, materials, resources, suppliers, project costs, reservations, goods receiving, documents, users, roles and permissions, with accounting, HR, equipment and analytics planned as further expansion areas.

The architecture documents define ConstructPro as a PHP MVC system using MySQL, Bootstrap 5, JavaScript/AJAX, jQuery, Select2 and FontAwesome, with an MVC + Services architecture.

2. PURPOSE

The primary purpose of ConstructPro is to provide a single operational system for controlling construction projects and the resources, materials, procurement activities and costs associated with them.

- Centralize construction project information.
- Control materials and stock across warehouses and locations.
- Connect procurement activities with receiving and supplier records.
- Track material reservations and issues to projects.
- Record and control project costs.
- Provide structured workflows instead of isolated transactions.
- Control access through users, roles and permissions.
- Create a foundation that can be extended into finance, HR, equipment and business intelligence.

3. CORE SYSTEM CONCEPT

ConstructPro separates master data, project activities, warehouse stock, transactions and workflow processing. A project can have costs, documents, quotations, procurement, materials and reservations, but it does not own physical inventory. Inventory belongs to warehouses and locations.

This separation allows the same inventory item to exist globally while its quantities are maintained by warehouse/location stock.

4. MAIN FUNCTIONAL AREAS

AREA	PRIMARY PURPOSE
Projects	Manage construction projects, project information, costs, documents, budgets, milestones and progress.
Customers	Maintain customer master

	information and connect customers with business activities and projects.
Inventory	Manage inventory master data, SKUs, categories, brands, units, stock and inventory history.
Warehouses & Locations	Manage physical storage locations and the stock held at each location.
Procurement	Manage purchasing activities from requests/RFQs through quotations, purchase orders and receiving.
Suppliers	Maintain supplier information, quotations, receipts, supplier ledger activity and payments.
Resource Requisitions	Request and control non-material resources and services required by projects.
Project Costs	Record project expenditure by cost type and connect material costs with inventory movements.
Reservations	Reserve material for a project or operational source before it is issued.
Goods Receiving & Returns	Receive purchased materials into stock and process returns while maintaining stock history.
Users, Roles & Permissions	Control system access, user accounts and permission-based functionality.
Documents & Printing	Produce operational documents and printable records for business workflows.
Future Finance	Customer ledger, expenses, payments, accounting and financial reporting.
Future HR	Employees, attendance, payroll, leave and related HR functions.
Future Equipment & Assets	Machines, vehicles, tools, fuel, maintenance and asset management.
Future Analytics	Dashboards, KPIs, forecasting, cash flow and project profitability.

5. PROJECT MANAGEMENT

The Projects module is the central operational context for construction work. The development specification identifies Projects, Project Costs, Documents, Budget, Milestones and Progress as core project capabilities, with Gantt, WBS, Resources and Scheduling identified as future extensions.

- Project master information and status.
- Project-related documents.
- Project quotations and procurement references.
- Project material requirements and reservations.
- Project cost recording.
- Budget and progress management.
- Connection to warehouse/material workflows.

6. INVENTORY & WAREHOUSE MANAGEMENT

Inventory is one of the most important architectural foundations of ConstructPro. The system uses one global inventory item for a material and maintains its physical quantities by warehouse/location. The architecture defines `inventory_location_stock` as the source of truth for current stock.

- Inventory master records.
- SKU and item identification.
- Categories, brands and units of measure.
- Warehouses and inventory locations.
- Location-specific stock quantities.
- Stock transfers between locations.
- Stock reservations.
- Inventory adjustments.
- Inventory movement history.
- Barcode/SKU-based item identification in operational workflows.

Stock changes are intended to occur through controlled stock operations such as `adjustStock()` and `transferStock()`, with an inventory movement created for every stock operation.

7. PROCUREMENT & PURCHASING

ConstructPro provides an integrated procurement flow connecting purchasing requirements with supplier quotations, purchase orders, goods receiving and supplier financial records.

The documented procurement workflow is:

1. Purchase Request
2. RFQ
3. Quotation
4. Purchase Order

5. Goods Receipt
6. Supplier Ledger
7. Invoice
8. Payment

This workflow is defined in both the architecture and development specifications.

8. SUPPLIERS & SUPPLIER MANAGEMENT

- Supplier master information.
- Supplier quotations and quotation comparison.
- Purchase orders connected to suppliers.
- Goods receipts against procurement.
- Supplier ledger activity.
- Supplier payments.
- Future supplier evaluation capabilities.

Supplier evaluation is identified as part of the procurement roadmap.

9. RESOURCES & REQUISITIONS

The system distinguishes material inventory from other resources and services. Resource requisitions provide a controlled way for projects or departments to request resources that may include labor, services, transport, equipment or other non-material requirements.

This distinction is important because only material project costs affect warehouse inventory. Labor and other non-material cost types do not directly change stock.

10. PROJECT COST MANAGEMENT

Project costs connect operational consumption and expenditure to a project. The documented cost types include Materials, Labor, Transport, Subcontract and Miscellaneous. Only Materials affect inventory.

- Material costs can be linked to warehouse stock.
- Material issues can create inventory movements.
- Labor and service-type costs remain financial/project costs without stock deduction.
- Project cost workflows can support project financial analysis.
- Future development can extend this foundation into project profitability and cash-flow reporting.

11. INVENTORY RESERVATIONS

Reservations provide a controlled mechanism for allocating available material to a project or another operational source without immediately reducing physical stock.

The documented workflow is Reserve → Fulfill. Reservation itself does not create a stock movement; fulfillment creates the corresponding OUT movement.

- Reserve material.

- Identify the source/project requiring the material.
- Fulfill the reservation when material is actually issued.
- Record the resulting inventory movement.
- Support cancellation or reversal workflows where applicable.

12. GOODS RECEIVING & RETURNS

Goods receiving connects purchase orders with warehouse stock. A receipt records the physical arrival of materials and increases the appropriate warehouse/location stock while creating an inventory movement.

The architecture specifies Goods Receipt as a workflow involving receipt, stock adjustment, movement creation and downstream supplier records.

- Receive against procurement.
- Update warehouse/location stock.
- Create inventory movement history.
- Maintain traceability to the purchasing process.
- Support goods return processing where required.

13. USERS, ROLES & PERMISSIONS

ConstructPro includes authentication, authorization, roles and permissions as core security capabilities. Warehouse access and project access are part of the security design, with approval levels and digital signatures identified as future enhancements.

- User accounts.
- User profile information and photo.
- Role assignment.
- Permission-based access.
- Administrative user management.
- Controlled access to sensitive operational functions.

14. SYSTEM ARCHITECTURE

ConstructPro follows an MVC + Services architecture. The intended flow is Controller → Service → Models → Database. Controllers receive and validate requests, services coordinate business workflows and transactions, and models handle database operations.

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- Controllers: request handling, validation, service invocation and redirection.
- Services: business rules, workflows, validation, transactions and rollback.
- Models: database SELECT, INSERT, UPDATE and DELETE operations.
- Views: user interface and presentation.
- Helpers/Core: shared system utilities and infrastructure.

This architecture is designed to prevent business logic from being scattered across controllers and models.

15. DATA & AUDIT PRINCIPLES

- Use structured master data rather than duplicating operational entities.
- Maintain timestamps and user attribution where applicable.
- Use explicit status values for workflow records.
- Maintain inventory movement history for stock operations.
- Keep transactions traceable to their originating business document.
- Use database transactions inside services for multi-step workflows.

The development specification calls for created/updated timestamps and user attribution where applicable, and identifies standard workflow statuses such as DRAFT, PENDING, APPROVED, REJECTED, CANCELLED and COMPLETED.

16. USER INTERFACE

ConstructPro uses a Bootstrap-based interface with tables, cards, forms, Select2 controls, modals, breadcrumbs and icons. The system also supports bilingual English/Arabic presentation, including RTL presentation for Arabic.

- Responsive Bootstrap 5 interface.
- English and Arabic interface support.
- RTL support for Arabic.
- Searchable selection fields.
- Operational tables and forms.
- Document-oriented screens for business workflows.
- User profile/photo presentation.

The development specification defines Bootstrap 5, cards, tables, Select2, modals, breadcrumbs and icons as UI standards.

17. INTEGRATED BUSINESS WORKFLOWS

17.1 PROCUREMENT TO STOCK

9. Requirement is identified.
10. Purchase Request/RFQ process is initiated.
11. Supplier quotations are received and evaluated.
12. Purchase Order is issued.
13. Goods are received.
14. Warehouse stock is increased.
15. Inventory movement is recorded.
16. Supplier ledger is updated.

17.2 PROJECT MATERIAL TO COST

17. Project requires material.
18. Material is identified and/or reserved.
19. Material is issued from the appropriate warehouse/location.
20. Stock is reduced through the controlled inventory operation.
21. Inventory movement is recorded.

22. Material consumption becomes a project cost.

The documented Project Engine follows Project → Budget → Resource Requisition → Issue Material → Project Cost → Reports.

17.3 WAREHOUSE TRANSFER

23. Source warehouse/location is selected.

24. Destination warehouse/location is selected.

25. Stock is transferred.

26. OUT movement is recorded for the source.

27. IN movement is recorded for the destination.

This movement-based approach is explicitly defined by the system architecture.

18. DOCUMENTS & OPERATIONAL RECORDS

- Project documents.
- Quotations.
- Purchase orders.
- Goods receipts.
- Goods returns.
- Resource requisitions.
- Inventory transfer records.
- Inventory reservation records.
- Project cost records.
- Supplier and financial records.

The system is intended to make operational records traceable through their related projects, suppliers, warehouses and transactions.

19. CURRENT DEVELOPMENT & ROADMAP

The supplied architecture identifies the current foundation around Inventory, Warehouses, Projects, Costs, Reservations, Transfers and Goods Receipt, followed by procurement, equipment/assets, accounting, HR and analytics.

PHASE	DIRECTION
Foundation	Authentication, users, roles, permissions and core system foundation.
Inventory	Inventory master, warehouses, transfers, reservations and movements.
Procurement	Purchase orders, RFQs, supplier evaluation and supplier ledger.
Equipment & Assets	Machines, vehicles, tools, fuel, usage and maintenance.
Accounting	Chart of accounts, journals, GL, trial balance, balance sheet and

	P&L.
HR	Employees, attendance, payroll and related HR functions.
Analytics	BI dashboards, KPIs, forecasting, cash flow and project profitability.

20. FUTURE EXPANSION

- Equipment and machine management.
- Fleet and vehicle management.
- Fuel management.
- Maintenance management.
- Asset management.
- Human resources, employees, attendance and payroll.
- Accounting and general ledger.
- Customer ledger, expenses and payments.
- BI dashboards and KPIs.
- Forecasting and project profitability.
- Mobile application.
- Offline synchronization.
- Barcode and QR-code workflows.
- GPS integration.
- Power BI integration.
- API expansion.

These future directions are based on the supplied architecture and development roadmap.

21. WHAT MAKES CONSTRUCTPRO DIFFERENT

- Construction-first design rather than a generic accounting system.
- Project-centered operational workflows.
- Warehouse-based inventory control.
- Movement-based stock history and traceability.
- Clear separation between physical material and other project costs.
- Integrated procurement-to-receipt-to-stock flow.
- Reservation and fulfillment logic for project material allocation.
- Permission-driven access control.
- Service-oriented workflow architecture prepared for future growth.
- Modular foundation allowing Finance, HR, Equipment and Analytics to be added without redesigning the core.

22. INTENDED USERS

USER GROUP	TYPICAL USE
Management	Projects, costs, procurement,

	inventory visibility and future dashboards.
Project Managers	Projects, materials, requisitions, reservations, costs and progress.
Procurement Staff	Suppliers, RFQs, quotations, purchase orders and receiving.
Warehouse Staff	Stock, receiving, transfers, reservations, issues and movements.
Finance	Supplier/customer ledgers, costs, payments and future accounting.
Administrators	Users, roles, permissions and system configuration.
HR	Future employee, attendance, payroll and HR processes.
Operations	Resources, services, materials and operational requisitions.

23. SYSTEM VISION

ConstructPro is intended to evolve into a complete construction ERP in which project management, procurement, warehouse operations, resources, costs, finance, HR, equipment and analytics operate as connected parts of one system.

The architecture deliberately establishes a stable foundation first: centralized inventory, controlled stock movements, project-centered cost tracking, structured procurement workflows, service-based business logic and permission-based access. This foundation is intended to make future modules additions to the ERP rather than separate systems.

The development specification is maintained as a living document, while the architecture document defines the fundamental design principles.